EUROPEAN WARS: FROM SAIL TO STEAM

DURING THE PERIOD from 1850 to 1870, navies struggled to adapt their tactics as the switch was made to steam power, ironclad ships, and rifled guns firing explosive shells. There were few chances for the European powers to experiment with new technology in action. In 1854 Britain and France went to war against Russia in support of Ottoman Turkey. Although known as the Crimean War, this conflict was fought in the Baltic as well as the Black Sea. Confronted by the world's two leading naval powers, Russia refused to commit its fleet to battle. Anglo-French command of the sea was essential to their land operations, but the allies had less success than hoped with the naval bombardment of land fortifications. In the 1860s Prussian ambitions to rule Germany and Italian aspirations to nationhood provoked a series of wars in which sea power was marginal. Nonetheless there were a couple of intriguing naval battles. The encounter at Heligoland was the last significant conflict between wooden ships, and the battle between Austrian and Italian fleets at Lissa was a remarkable instance of the brief fashion for ramming as the prime means of attack.

CRIMEAN WAR

SINOPE



Date November 30, 1853
Forces Russians: 11 ships; Ottoman Turkish: 12 ships
Losses Russians: none; Ottoman Turkish: 11 ships sunk

Location Sinope, northern Turkey

A month after the Russian and Ottoman Empires had gone to war, a powerful Russian naval squadron under Admiral Pavel Nakhimov attacked an Ottoman force of seven frigates and five smaller ships in harbor at Sinope. Nakhimov's squadron included six battleships and mounted a total of 720 guns, some of them the new Paixhans guns that fired explosive shells. This overwhelming weight of firepower devastated the Turkish ships in an hour, in a battle fought at anchor. Only one Ottoman vessel escaped destruction, a steam-powered auxiliary under an English captain. British and French public opinion responded to the attack (a legitimate act of war) with outrage. The battle was denounced in the press as the "massacre at Sinope" and offered the pretext for Britain and France to enter the war against Russia.



Exploding shells
Nakhimov's warships open fire on the Turkish fleet in an action that, with its use of exploding shells, saw an end to the era of unarmored ships.

SECOND SCHLESWIG WAR

HELIGOLAND



Date May 9, 1864
Forces Danish: 2 frigates, 1 corvette; Prussians and Austrians: 2 frigates, 3 gunboats
Losses Danish: none; Prussians and Austrians: 1 frigate badly damaged

Location Off Heligoland, North Sea

In 1864 Denmark fought Prussia and Austria over control of the duchies of Schleswig and Holstein. The Danes, who were superior to the Prussians at sea, blockaded ports in northern Germany with a squadron under Commodore Edouard Suenson. Captain Wilhelm von Tegetthoff took a small Austrian squadron to the North Sea to break the blockade. It encountered Suenson's force on May 9 off the island of Heligoland, then a British possession. Each side had two wooden-hulled steam-powered screw frigates: the Danish *Niels Juel* and *Jylland*, and the Austrian *Schwarzenberg* and *Radetzky*. The Danish steam corvette *Heimdall* also took part in the fighting. The hostile squadrons first steamed past one another on opposite tacks shelling furiously, then maneuvered to follow a parallel course, Suenson's *Niels Juel* taking on Tegetthoff's *Schwarzenberg* while *Jylland* and *Heimdall* engaged *Radetzky*. After more than two hours' fighting, *Schwarzenberg* burst into flames. Unable to continue fighting, Tegetthoff fled for neutral British territorial waters around Heligoland. The Danes had also suffered damage, especially to *Jylland*, and withdrew for repairs. Both sides claimed a victory, the Danes having inflicted heavier casualties but the Austrians having succeeded in lifting the blockade.

CRIMEAN WAR

SEVASTOPOL FORTS



Date October 17, 1854
Forces British and French: 27 ships
Losses British and French: 340 casualties; Russians: 1,100 casualties

Location Sevastopol, Crimea

The siege of the Russian naval base of Sevastopol was the gruesome centerpiece of the Crimean War. The British and French fleets landed some 50,000 troops in September 1854 to besiege the city on land. The sea approach to Sevastopol Harbor ran up a narrow creek, the mouth of which was guarded by forts Constantine and Alexander and blocked by five ships of the line and two frigates sunk there by the Russians. On October 17 a combined land and sea attack on Sevastopol was attempted. The task of the British and French fleets, under Vice Admiral James Dundas and Vice Admiral Ferdinand Hamelin, was to destroy the forts at the harbor mouth. They deployed a mix of steam and sailing ships of the line, the latter moved into position by small steamers used as tugs. In a long-range exchange of fire with the shore batteries, the unarmored wooden ships proved excessively vulnerable to the Russians' shells and red-hot shot, several being forced to leave the line. Fort Constantine suffered severe damage when a magazine exploded, but the shore batteries were still firing when the Allied fleet withdrew to lick its wounds. The long siege lasted until September 1855, when Sevastopol was finally taken by Allied land forces after heavy loss of life.

Sevastopol besieged
A map of the siege of Sevastopol as it was on June 18, 1855. The Allied fleet enters the harbor to the north, while Russians flee from ruined fortifications.



Russian naval sword
Though originally designed as a boarding weapon, by 1860 the naval sword was only used when going ashore or on ceremonial occasions.